Water Quality Prediction in Arizona Using Gaussian Process Regression

A Gaussian Process Regression System for Water Quality Assessment Using FAAS Field Data

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Engineering Problem

Dissolved metals (Fe, Cr, Mn, Mo, In) in urban water bodies indicate contamination from industrial runoff, aging infrastructure, and natural geology.

Several metals are regulated under EPA National Primary and Secondary Drinking Water Standards (e.g., Cr MCL = 0.1 mg/L; Fe Secondary MCL = 0.3 mg/L).

Laboratory analysis costs **$50–$200 per sample** and takes days to return results.

The Phoenix metro area spans **thousands of square kilometers** with dozens of lakes, rivers, and canals.

Environmental agencies cannot afford to test every water body - resources must be prioritized.

Engineering Goal

Design and deploy a spatial prediction system that estimates dissolved metal concentrations at **unsampled locations** using sparse FAAS field data, with built-in uncertainty quantification to guide future sampling priorities.

Desired Improvement

- Reduce cost of spatial water quality assessment
- Provide actionable uncertainty maps that identify where additional sampling is most needed

Existing Solutions and Limitations

Direct Sampling + FAAS
Gold standard for accuracy, but prohibitively expensive at scale ($50–$200/sample, days per result). Covers only tested locations.

Inverse Distance Weighting (IDW)
Simple spatial interpolation; assigns no uncertainty and assumes uniform spatial behavior.

Kriging (Geostatistics)
Provides uncertainty estimates, but standard implementations require dozens to hundreds of observations for reliable variogram fitting.

Neural Networks
Powerful but require large labeled datasets (thousands of samples); opaque predictions with no built-in confidence measure.

Gap This Project Addresses

No existing low-cost method produces **both predictions and calibrated uncertainty** from **extremely sparse data** (n = 6). Gaussian Process regression is mathematically suited to small datasets and provides **native posterior uncertainty** at every prediction point. No prior publicly available GP-based water quality tool exists for the Phoenix metropolitan area.

Engineering Design Concept

Analytical Method: Flame Atomic Absorption Spectroscopy (FAAS)

Measures absorption of element-specific light by ground-state atoms in an air–acetylene flame (~2300 °C). Follows the Beer–Lambert Law: $\mathbf{A = \varepsilon \cdot l \cdot c}$

Triplicate readings per sample per metal; concentrations determined from calibration curves ($R^2 \geq 0.994$).

Core Algorithm: Gaussian Process (GP) Regression

A GP defines a distribution over functions; it finds the smoothest surface passing through observed data points. Predictions are weighted averages of known values, with weights determined by a **kernel function** that encodes spatial correlation.

System Architecture

FAAS measurement → 5 metals × 6 sites × 3 replicates = 90 triplicate readings

5 independent GP models - one per metal analyte (Fe, Cr, Mn, Mo, In)

Composite RBF kernel: short-range component (optimized) + long-range 150 km component (fixed) captures both local variation and regional trends

Hyperparameter optimization: grid search over 288 combinations per metal, maximizing log marginal likelihood

Real-time web deployment: React frontend + backend function serving GP predictions on demand

Why This Design Works

- GPs are mathematically optimal for small-sample spatial interpolation
- The composite kernel captures multi-scale spatial structure
- Built-in uncertainty grows with distance from data, providing honest confidence bounds

Prototype Development

1

v1 - Single RBF Kernel
Under fitted regional trends; predictions collapsed to mean beyond ~20 km.

2

v2 - Composite Kernel
Short + long-range RBF: captured both local variation and broad spatial trend.

3

v3 - Full FAAS + Uncertainty Pipeline
Added FAAS calibration workflow, posterior variance computation, extrapolation flagging, and LOOCV validation.

Engineering Trade-offs

Trade-off	Decision	Rationale
Accuracy vs. computational cost	Grid search (288 combos) instead of gradient optimization	Exhaustive search is feasible with 6 training points; avoids local minima
Model complexity vs. interpretability	Independent GP per metal instead of multi-output GP	Multi-output requires more data to learn cross-covariance reliably
Spatial resolution vs. data availability	Equirectangular km projection instead of raw lat/lon	Makes length-scale physically meaningful (km, not degrees)
Prediction honesty vs. coverage	40 km extrapolation threshold with explicit flagging	Prevents false confidence in distant predictions
Analytical precision vs. throughput	Triplicate FAAS readings per sample	Balances measurement uncertainty quantification with lab time

Implementation

Analytical: FAAS with element-specific HCLs (Fe 248.3 nm, Cr 357.9 nm, Mn 279.5 nm, Mo 313.3 nm, In 303.9 nm). Backend: Deno/TypeScript edge function with from-scratch GP implementation (Cholesky decomposition, forward/back substitution). Frontend: React + Google Maps with interactive click-to-predict interface. Numerical stability: jitter term (10^{-10}) added to kernel diagonal; Jacobi eigenvalue verification.

Experimental Setup

Analytical Method

Flame Atomic Absorption Spectroscopy (FAAS), air–acetylene flame. Five element-specific hollow cathode lamps. Five-point calibration per metal from certified 1000 mg/L stock solutions.

Sampling Sites

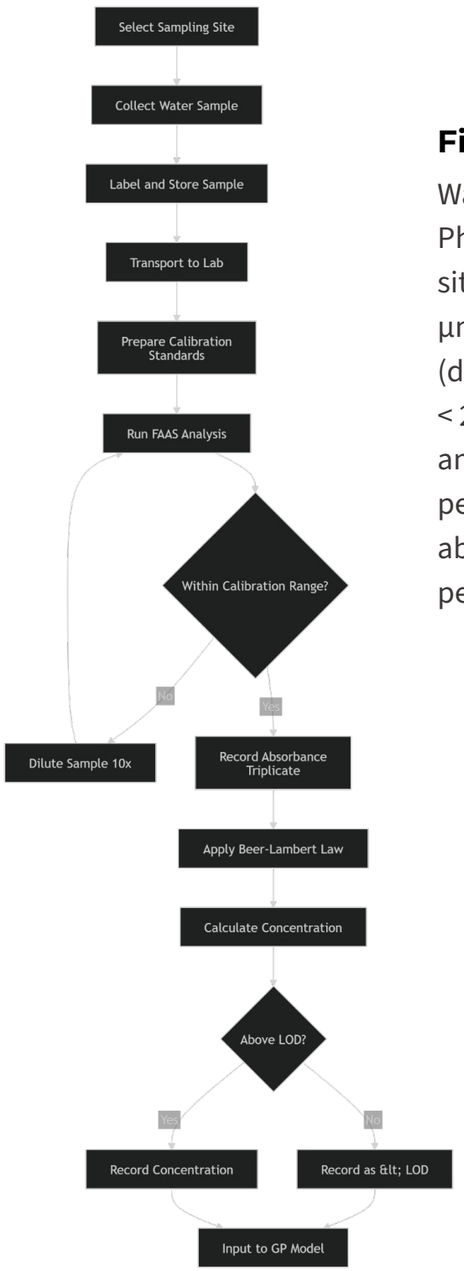
Six water sampling sites across metropolitan Phoenix, spanning ~14 km (N–S) × ~34 km (E–W). Sites: Alvord Lake, Desert West Lake, Steele Park, Papago Park, Gila River, Tres Rios.

Calibration Equations

Metal	Calibration Equation	R^2	Wavelength (nm)
Fe	A = 0.008c + 0.002	0.997	248.3
Cr	A = 0.042c + 0.003	0.998	357.9
Mn	A = 0.038c + 0.002	0.996	279.5
Mo	A = 0.43c + 0.001	0.995	313.3
In	A = 0.25c + 0.002	0.994	303.9

Validation Method: Leave-One-Out Cross Validation (LOOCV)

For each of the 6 sites: remove it, retrain the GP on the remaining 5 (including full hyperparameter re-optimization), predict the held-out site. LOOCV is the most rigorous validation possible with n = 6 - ensures zero data leakage. Total: 6 folds × 5 metals = 30 individual predictions evaluated.



Data Collection

Field Measurements

Water samples collected from 6 Phoenix-area sites (3 replicates per site). Samples filtered through 0.45 µm cellulose acetate filters (dissolved fraction). Acidified to pH < 2 with trace-metal-grade HNO_3 and transported at < 6 °C. Analysis performed by FAAS with triplicate absorbance readings per sample per metal.

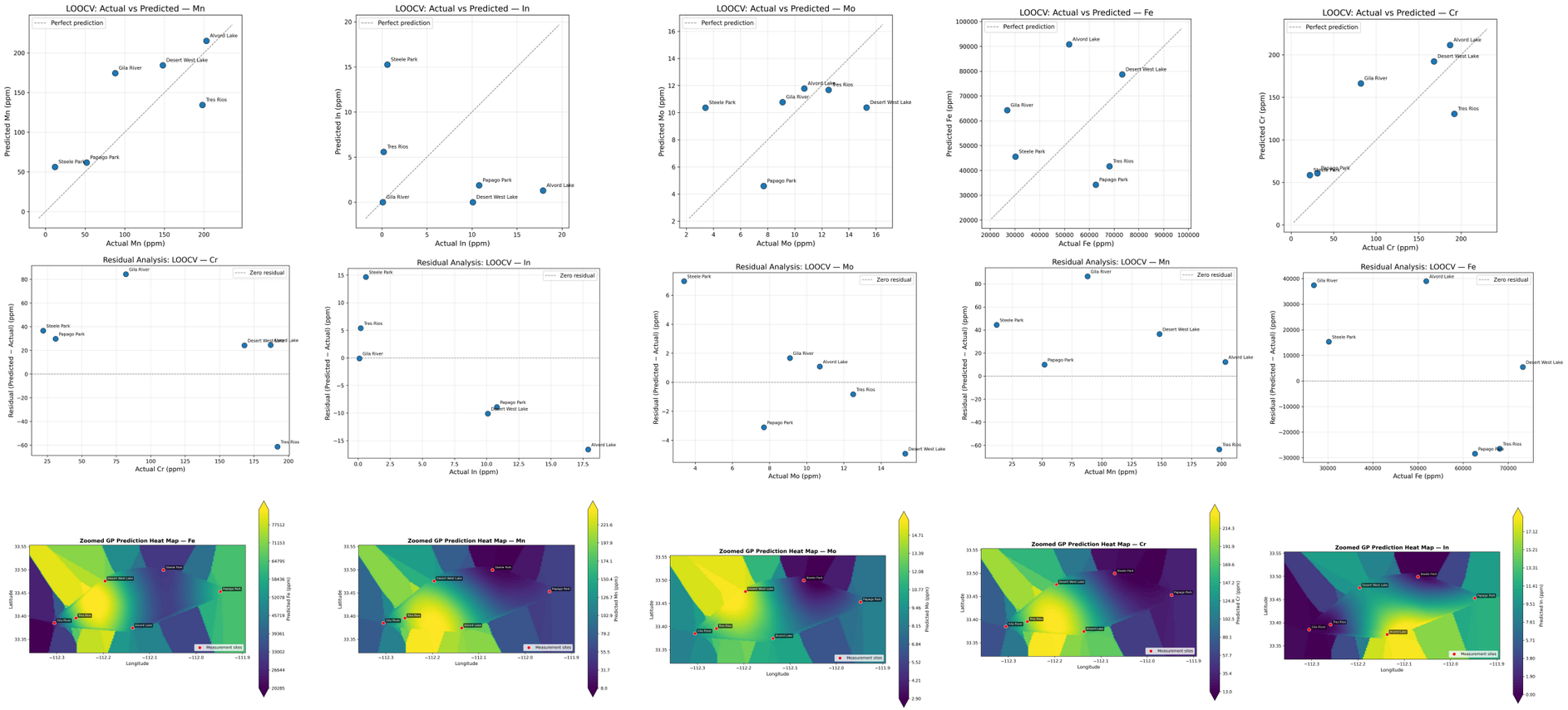
Measured Concentrations (site means)

Site	Fe (mg/L)	Cr (mg/L)	Mn (mg/L)	Mo (mg/L)	In (mg/L)
Alvord Lake	51.84	0.188	0.206	0.0105	0.0182
Desert West Lake	73.17	0.167	0.149	0.0157	0.0102
Steele Park	30.12	0.019	0.012	0.0033	< LOD
Papago Park	62.66	0.031	0.048	0.0085	0.0107
Gila River	26.86	0.088	0.090	0.0100	< LOD
Tres Rios	68.16	0.198	0.203	0.0126	< LOD

Key Observations

- Fe required 10× dilution at all sites (concentrations exceeded linear calibration range)
- In fell below the limit of detection at Steele Park, Gila River, and Tres Rios
- Cr and Mn highest at western sites (Alvord Lake, Tres Rios); lowest at eastern sites (Steele Park, Papago Park)
- Concentration ranges: Fe 26.9–73.2 mg/L (2.7×), Cr 0.019–0.198 mg/L (10.4×), Mn 0.012–0.206 mg/L (17.2×)

Results



Key Quantitative Findings

- Cr and Mn exhibited the **strongest spatial structure** - clear west–east concentration gradient with predictions closely tracking held-out measurements in LOOCV.
- Cr, Mn, and Mo **co-varied**, suggesting a common industrial or geological source in western Phoenix (Alvord Lake, Tres Rios, Desert West Lake).
- Fe had the **largest absolute RMSE** (~30,000 ppm), consistent with its wide concentration range.
- Predictive uncertainty **increased monotonically** with distance from training sites. Predictions beyond **40 km** automatically flagged as extrapolations.
- Hyperparameter optimization selected distinct kernel parameters per metal, confirming each metal has different spatial behavior.
- In was the **least predictable** - below LOD at half the sites, no clear spatial gradient.

Interpretation of Results

How Well the System Met the Engineering Goal

The GP system successfully produces **spatially smooth, physically reasonable predictions** at unsampled locations within the Phoenix metro area. Every prediction includes a **calibrated uncertainty estimate** (posterior standard deviation) that grows with distance from training data. The **extrapolation flag** (> 40 km) prevents overconfidence in distant predictions.

Weakest Performance

Fe: Large absolute errors due to enormous value range (26.9–73.2 mg/L), but the model still identifies relative spatial trends.
In (Indium): Below LOD at 3 of 6 sites; detectable values (0.010–0.019 mg/L) show no clear spatial gradient - insufficient data for reliable GP modeling.

Strongest Performance

Cr and Mn: Clear west–east gradients confirmed by both FAAS measurements and GP spatial modeling. Highest concentrations at Alvord Lake (Cr: 0.188, Mn: 0.206 mg/L) and Tres Rios (Cr: 0.198, Mn: 0.203 mg/L); lowest at Steele Park (Cr: 0.019, Mn: 0.012 mg/L).

Metal co-variation: Cr, Mn, and Mo co-vary across sites - consistent with shared industrial/wastewater sources in western Phoenix near Tres Rios water reclamation facility.

Comparison to Expectations

- Hypothesis confirmed: predictions are reasonable nearby, degrade with distance, and uncertainty correctly reflects degradation
- West–east Cr/Mn gradient hypothesis confirmed by FAAS measurements
- In performance was weaker than expected due to LOD constraints

Limitations, Improvements, Conclusions & Applications

Limitations

Extremely small dataset (n = 6): The smallest practical size for GP regression; limits the complexity of spatial patterns the model can learn

All sites within metro Phoenix: No training data outside the ~14 × 34 km cluster; all distant predictions are extrapolations

LOD constraints for In: Indium below detection limit at 50% of sites eliminates meaningful spatial structure for this analyte

No temporal dimension: Concentrations change seasonally; this model captures only a sing snapshot

2D spatial inputs only: No geology, land use, upstream sources, or water chemistry features included

Isotropic kernel assumption: The model assumes spatial correlation is the same in all directions, which may not hold near rivers or industrial corridors

FAAS limitation: FAAS measures total elemental concentration, not speciation (e.g., cannot distinguish Cr(III) from Cr(VI))

Planned Improvements

Increase sample size: Adding 10–15 sites across the greater Phoenix area would dramatically improve LOOCV performance and spatial coverage

Temporal modeling: Collect seasonal data to build a spatio-temporal GP

Replace In with a more detectable analyte or use a more sensitive technique (ICP-MS) for trace-level metals

Feature enrichment: Incorporate land-use type, proximity to industrial zones, and upstream distance as additional input features

Anisotropic kernel: Allow different length scales in different spatial directions to capture river-corridor effects

Multi-output GP: With more data, a joint model could leverage Cr–Mn correlation to improve predictions for both metals

Chromium speciation: Pair FAAS with colorimetric Cr(VI) assay to distinguish toxic hexavalent chromium

Key Findings

FAAS analysis of six Phoenix-area water bodies quantified dissolved concentrations of five metals with triplicate precision ($R^2 \geq 0.994$ for all calibration curves)

Cr and Mn show the strongest predictable west–east spatial gradient, with concentrations ~10× higher at western sites (Alvord Lake, Tres Rios) than eastern sites (Steele Park)

Cr, Mn, and Mo co-vary, suggesting a common contamination source in western Phoenix

A Gaussian Process trained on these **6 FAAS measurements** produces physically reasonable spatial predictions with honest, calibrated uncertainty

The **composite kernel** (short-range + 150 km long-range) outperforms a single RBF by capturing multi-scale spatial variation

In is not reliably predictable at these concentration levels with FAAS (below LOD at 50% of sites)

Engineering Goal Achieved

The system delivers real-time, interactive spatial predictions with honest confidence bounds - deployed as a functional web application.

Applications

Environmental monitoring: Agencies can use uncertainty maps to prioritize where to collect the next water samples - targeting high-uncertainty regions for maximum information gain

Resource allocation: Reduces the number of expensive lab tests needed to characterize a metropolitan water system

Urban planning: Identifies potential contamination hotspots (e.g., elevated Cr/Mn clusters near Tres Rios and Alvord Lake in west Phoenix) for further investigation

Scalable framework: The same FAAS + GP pipeline can be applied to other cities, other pollutants, or other sparse-data spatial problems

Future Directions

- Expand sampling network to 15–20 sites
- Add seasonal sampling for temporal dimension
- Incorporate land-use and geological data as input features

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